

Forecasting Inflation Using Google Trends Data

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Abstract

This paper examines the addition of Google Trends keyword search data as a measure to reduce inflation forecast errors and improve accuracy in high inflation countries to mitigate data publishing issues. We focus on developing countries with high internet penetration and Consumer Price Index (CPI) availability in the Middle East and South America, namely Iran, Jordan, Egypt, Suriname, Brazil, Uruguay, and Argentina. A growing literature demonstrates the effectiveness of using Google Trends data in improving inflation forecasts. We address a gap in the literature as it pertains to our chosen countries given their high inflation statuses. We select annual and monthly CPI data and Google Trends keywords from 2004 to 2024, categorizing the keywords into four channels: CPI, Currency and Investment, Labor Market, and Financial Stress. We use the classical Auto-Regressive (AR) and Auto-Regressive Integrated Moving Average (ARIMA) forecasting models with 14 (2004-2017) periods of training and 7 (2018-2024) periods of test data. Our analysis results confirm expected findings that Google Trends data improves inflation forecast with errors reduced by up to 56% in Jordan (annual data) and 8.6% in Egypt (monthly data) compared to baseline models excluding Google Trends.

JEL Codes:

E31 Price Level – Inflation – Deflation

E37 Forecasting and Simulation: Models and Applications.

Keywords: Inflation Rate, Forecasting, Google Trends, CPI, Middle East, South America, Iran, Jordan, Egypt, Suriname, Brazil, Uruguay, Argentina, Internet Penetration

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I Introduction

Inflation is a vital macroeconomic indicator that directly impacts the health and security of a given economy via interest rates, investment, balance of trade, and Gross Domestic Product (GDP). Monetary and fiscal policy decisions help drive inflation, demonstrating how past decisions impact the present. However, macroeconomic reports are typically released with delays. Inflation forecasts therefore encounter a variety of difficulties in improving their accuracy. Obtaining the most up-to-date information would ideally help mitigate this issue as it pertains to inflation forecasts.

To address this issue, complementary data sources such as Google Trends data can capture search behavior in real time and might reflect consumer sentiments and household perceptions as a proxy before official macroeconomic reports release with actual and forecasted inflation rates. If searches for unemployment, dollar, or inflation rate are increasing, this might indicate some accompanying increase in expectation or popularity of those ideas or related topics. If someone is worried about increasing gas prices because of a recent global conflict, they might search on Google for gas prices and then for cheapest gas prices near them. However, Google Trends data does not provide a direct or unbiased measure of consumer sentiment, as search activity can be driven by media attention, heterogeneous internet access, and curiosity rather than genuine economic concern. Nevertheless, the Search Volume Index (SVI) captures a wider set of related queries linked to a given term, enabling it to proxy changes in public attention and perceived economic stress in near real time, often ahead of official data releases.

The primary objective of this research is whether the addition of Google Trends data improves inflation forecasts with a focus on South American and Middle Eastern countries. We evaluate whether Google Trends data is a compatible data source for inflation forecasting by adding specific keyword search volumes with the goal of capturing consumer sentiment related to inflation expectations. Using other macroeconomic variables such as money supply, Consumer Price Index (CPI), exchange rates, global oil price (Brent crude oil) and unemployment rate, we compare base models (predicting inflation using lagged values of inflation) with augmented Google Trends models to see if the inclusion of inflation-related search information helps improve inflation forecasts. All our selected countries exhibit volatile inflation rates due to unstable economic conditions, policy, or international conflict. We hypothesize that the addition of Google Trends data to our models will more closely capture consumer inflation expectations with a shorter time lag between government inflation reports and actual inflation numbers. Given that our seven countries exhibit high inflation and unstable economic environments, this research seeks to provide a framework for other countries with similar conditions. Improving inflation forecasts with Google Trends data would be a major asset to economists and policymakers to stabilize their economies and create sound policy.

II Literature Review

Numerous studies have examined whether Google Trends data is helpful in forecasting macroeconomic indicators such as GDP and inflation across different countries. The general finding across the literature is that incorporating Google Trends data improves inflation forecasts and enhances predictive accuracy compared to models that rely solely on traditional macroeconomic variables.

Petropoulos et al., (2022) focuses on creating a composite keyword index using Google Trends search volume. In this study, the authors construct a financial term dictionary using text mining techniques such as TF-IDF and LDA applied to over 10,000 central bank speeches. They test whether search intensity for these financially relevant terms can predict a crisis one month ahead. Using machine learning and deep learning models they find that Google search activity contains predictive information, with deep neural networks outperforming traditional statistical models. Earlier work by Giselle Guzmán (2011) similarly highlights the forecasting utility of internet search behavior by constructing a Google Inflation Search Index (GISI) showing that search-based measures can outperform conventional low-frequency expectations indicators in predictive accuracy and Granger Causality tests, suggesting that higher-frequency search metadata can meaningfully capture inflation expectations. Complementing these findings, Bleher and Dimpfl (2022) developed a methodology for constructing a composite index from Google Search Volume Index (SVI) data, which overcomes Google's time-range and scaling limitations.

In a 2025 study by Roldán-Ferrín & Parra-Polanía, the authors integrate Google Trends with a Random Forest machine-learning framework to nowcast monthly inflation in Colombia, concluding that this hybrid model outperforms traditional statistical techniques (e.g., SARIMA, Ridge, Lasso) and rivals professional survey forecasts while producing predictions earlier in the data cycle. Their findings demonstrate the practical value of alternative data sources for emerging-market inflation monitoring. Krzysztof Drachal (2020) applies Dynamic Model Averaging (DMA) alongside Google Trends data to forecast inflation in the U.S. and Poland. He demonstrates that time-varying model combinations improve inflation forecasts and this conclusion illustrates how search queries are dynamic drivers of inflation across different economies. Similarly, Bantis, Clements, and Urquhart (2023) examined whether Google Trends data can improve GDP growth forecasts in the United States and Brazil relative to traditional macroeconomic forecasting methods. These papers show that incorporating Google Trends data alongside traditional macroeconomic indicators improves forecasting accuracy.

While there are studies proving how Google Trends data improves predictive accuracy, there are also studies which analyzed the usefulness of Google Trends search data by confining their search to specific keywords. One such research is Simionescu et al. (2020), where they explore unemployment during Brexit. Brexit refers to the United Kingdom's decision to leave the Euro-

pean Union following the 2016 United Kingdom European Union membership referendum. The vote exposed deep political and economic divisions, particularly around trade, immigration, and sovereignty. The formal exit in 2020 marked a major structural shift, introducing new trade barriers and regulatory frictions between the UK and EU. In the short run, Brexit created uncertainty, currency volatility, and investment slowdowns, while its long-term economic effects—especially on trade and productivity—are still unfolding. In this paper, the authors use search intensity for the keyword Brexit as an independent variable and find that a one unit increase in the Google Trends index is associated with a -0.029 percentage point change in unemployment.

Bikourne et al. (2025) made it their goal “to produce accurate monthly inflation forecasts in Morocco...to determine if a statistically significant relationship exists between Google Trends data and the inflation rate in real time.” (p. 2). Since common languages in Morocco are Arabic, French, and English, the authors selected 105 terms across the three languages and sorted them into categories: Energy, Finance, Food, Website, and Other Keywords. Some English language examples include *dollar*, *inflation*, *energy*, *product*, *tax*, *bank*, *money*, *oil*, *interest rate*, etc. This is of great interest to the structure of our research because it informs us that selecting keywords in the local language of a given country is a good place to start when constructing an index of search terms. Additionally, the authors selected the keywords based off of a volume or an increase in search traffic and sorting them into categorical bins might be helpful. They created a Google Trends index using the most pertinent keywords with a Principal Components Analysis (PCA) dimensionality reduction technique where the PCA selected 18 keywords containing the most relevant information from the original 105. Interestingly, *dollar* was no longer included in the 18 keywords after implementing PCA, but *oil* in English and its translations from French and Arabic. The index decreased multicollinearity and is helpful for flexibility if data must be “transformed into a single time series.” (p. 13). This study shows that including search terms from the local language are beneficial in the PCA index creation. The keyword selections in the PCA index suggest that other economic concerns such as energy are helpful in capturing inflation expectations.

Across the literature, the two main advantages of using Google Trends data are its flexibility and its nature of higher frequency data. Together, these studies indicate a robust role for internet search data combined with advanced econometric and machine-learning methods in enhancing the timeliness and accuracy of inflation forecasts.

III Data Discussion

A Macroeconomic Variables

This analysis uses monthly and annual Consumer Price Index (CPI) data for Iran, Jordan, Egypt, Suriname, Brazil, and Uruguay from the World Bank’s International Financial Statistics database and from FRED. Argentina’s CPI data was only available for 2017-2024 years, which is a very short range to analyze at an annual frequency and hence Argentina is only analyzed at the monthly frequency. The Appendix includes country-level summary statistics for each variable. The CPI datasets have 21 observations at the annual percentage level and 240 observations at the monthly level from 2004 to 2024. Other macroeconomic variables used from the World Bank are exchange rate between local currency units (LCU) per US dollar, M2-money supply, Brent crude oil prices, and unemployment rate. The dataset used, its source and its frequency is listed in Table 1.

B Google Trends Data

This research uses monthly and annual Google Trends data from 2004-2024. Table 1 lists the source and frequency of our datasets. We used the publicly accessible search volume index data from the Google Trends website with the *pytrends* package in Python. The primary limitation with using *pytrends* is that only five keywords may be downloaded at a time. The Search Volume Index (SVI) in Google Trends is normalized on a scale from 0 to 100, where 100 represents the highest observed search interest for a given query across the selected time period and region. When multiple keywords are included, the values are scaled relative to the peak search intensity among all terms, meaning that lower values reflect proportionally lower search interest rather than absolute differences in search volume. Without access to the proprietary Google Trends data, we are unable to determine the absolute number of searches performed for a keyword query. Testing revealed that there are no missing values across our Google Trends data sets. Available technical documentation did not attest to any imputations or measurement issues.

Table 1: Data Sources

Source	Data	Frequency	Time Period
World Bank	Inflation	Annual	2004–2024
World Bank	Exchange rate (LCU/USD), unemployment, and M2	Annual	2004–2024
World Bank	HCPI (harmonized CPI)	Monthly	2004–2024
IMF / EIA	Brent crude oil price	Annual / Monthly	2004–2024
Google Trends	Search Volume Index (SVI)	Annual / Monthly	2004–2024

C Composite Index Construction

To filter the data before running models, we performed tests on the merged Google Trends search volume index and CPI dataset for stationarity, correlation, Granger Causality, cross-correlations, and channel tests. We computed pearson correlation of each surviving series with year over year inflation using CPI. The series whose correlation was below the 0.2 threshold are dropped; the correlation filter is the only strict filter for selecting keywords that will be the most helpful in our analysis.

After filtering, we made three composite search volume indexes of the keywords to further delineate the optimal choice for inflation forecasting. Our three indexes are (1) simple average, (2) equal channel weight, correlation-weighted within channel, and (3) survival-rate channel weight, correlation-weighted within channel. The forecast combination literature shows simple equal-weight averages often outperform weighted combinations in small samples. Reporting all three lets us compare and ensure a more robust methodology.

IV Methodology

A Country Background

The aim of this paper is to find an alternate data source to produce accurate forecasts in those countries where data publishing is inconsistent. The seven countries chosen for this research focuses on exhibit fluctuating inflation. Due to unstable economic conditions and inconsistencies with release schedules, data such as GDP, inflation, CPI, and other economic variables are subject to large amounts of missingness or lack of reporting from respective government statistical agencies. This is an important caveat to keep in mind for inflation forecasting, thus we provide some background on each of the countries to better understand their economic conditions.

Uruguay represents a relatively stable inflation-targeting regime characterized by persistent but moderate inflation, largely shaped by wage indexation, exchange-rate pass-through. Historically, Uruguay's economy has been closely tied to its regional neighbors, Argentina and Brazil, with strong trade and financial linkages influencing its macroeconomic performance. However, this dependence has declined over time, as noted by The Economist (Mercopress, 2018). Uruguay has increasingly diversified its trade structure. Despite this reduced reliance, Uruguay remains highly exposed to external shocks, particularly due to its dependence on oil imports. Unlike other countries in the sample, Uruguay did not experience a systemic macroeconomic crisis but rather periods of deceleration associated with regional spillovers, followed by temporary inflationary pressures linked to global commodity shocks and drought. By 2024, Uruguay appears to be in a mature stabilization stage, with inflation increasingly anchored within target bands and institutional

credibility strengthening.

Suriname is a commodity-dependent nation where inflation is largely driven by external shocks and currency instability rather than conventional domestic demand factors. Suriname's economy depends heavily on mining exports such as gold and bauxite and is vulnerable to commodity price shocks (Ooft, 2019). Bauxite was the main driver of the economy from its independence from the Netherlands in 1975 to the early 2000s and then gold became a major export in 2004 (Ooft, 2019). Crude oil became another major sector of the Surinamese economy in the 1990s (Ooft, 2019). Due to fall in aluminium prices in 2015–2016 and no savings buffer, the nation experienced hyperinflationary episode and a subsequent debt and adjustment crisis. Towards the end of our sample data in 2021, Suriname remains in a prolonged adjustment stage under external support from the IMF. That year, the IMF approved \$688 million USD to Suriname over 36 months and an immediate disbursement of \$55.1 million USD to help with fiscal stability, bring down public debt, and even to fortify their "institutional capacity to tackle corruption and money laundering and improve governance" (IMF, 2021).

According to Nicolini and Kehoe (2022), Brazil has had a long history of high inflation pre-1994 due to passive monetary policy, fiscal deficits, and institutional changes. Accompanied with the election of President Lula, Brazil experienced economic growth from 2003 to 2011 in terms of higher GDP per capita, increased international trade, reduction of public debt, and invested in social welfare programs to combat income inequality. Economic growth slowed starting in 2012. From 2012 to 2016, they comment how the Brazilian economy and inflation rate are linked to commodity prices, government intervention in state-owned businesses, and policies that led to the deterioration of fiscal balances (Nicolini and Kehoe, 2022).

Egypt can be understood as a crisis-prone inflation regime dominated by exchange-rate pass-through, recurrent balance-of-payments pressures, and policy-induced price shocks. Egypt is heavily reliant on imports of food. In 2016, Egypt allowed its currency to float and it subsequently became devalued; imports suddenly became extremely expensive (Associated Press, 2016). Pure inflation models will struggle to anticipate this event. Egypt is heavily reliant on Russia and Ukraine to import wheat. The onset of the Russo-Ukrainian war (2022) raised import costs, triggering currency depreciation and higher inflation. The currency float and sharp increase in import costs represent two major structural break incidents that sharply altered inflation dynamics. By 2024, Egypt appears to be in a stage of externally supported stabilization, where crisis risks have been partially contained through IMF support and currency adjustment; inflationary pressures and macroeconomic imbalances remain elevated.

Jordan's economic conditions over 2004–2024 can be characterized as macroeconomic stability alongside persistent structural fragility. The economy experienced strong growth in the mid-2000s driven by liberalization and capital inflows, but this momentum was disrupted by external shocks,

including the 2008 global commodity price spike and the spillover effects of the Arab Spring, particularly through refugee inflows and regional instability. While Jordan avoided domestic political crisis, these pressures strained public finances, increased debt, and kept unemployment persistently high. Despite these challenges, inflation remained relatively low and stable due to the long-standing peg of the Jordanian dinar to the US dollar, which anchored price levels. Subsequent IMF-backed reforms, including subsidy cuts and tax increases, helped maintain fiscal stability but slowed growth and triggered social discontent. The COVID-19 shock further weakened the economy. By 2024, Jordan has stable inflation; however, ongoing vulnerabilities related to high unemployment, slow growth, and exposure to external shocks still remained. The status quo policies are insufficient to solve the aforementioned problems and hence the government needs to implement new policies to meet the growing populations' demands (Esrig, 2018).

The Iranian economy is historically sensitive to political pressures, economic sanctions, and the oil market with the Strait of Hormuz supplying 20% to 25% of the world's petroleum supply. According to the Brookings Institution, the 2004-2008 economy was marked with a housing bubble causing apartment costs to skyrocket, unemployment as high as 20% to 30%, and growing income inequality (Campbell, 2008). For example, the United States imposed oil sanctions on Iran in 2013 due to Iran's nuclear development program, plummeting the value of the rial. These sanctions heavily affected inflation in Iran (Clayton, 2025). In 2015, the United States and other countries entered into the Joint Comprehensive Plan of Action (JCPOA) to curtail Iran's nuclear development but allow some nuclear activities to take place; however, U.S. sanctions were still in place for Iran's support of various other activities such as missile development that conflicted with U.S. foreign policy standards (Clayton, 2025). After the U.S. pulled out of the JCPOA in 2018 under the Trump Administration, another instance of U.S. imposed sanctions on Iran occurred that year due to "continued malign activities" (Clayton, 2025). These sanctions substantially altered inflation dynamics in Iran. By 2024, Iran stays in a stage of chronic crisis management rather than stabilization, with inflationary pressures, currency instability, and institutional distortions continuing to dominate the macroeconomic environment.

Therefore, alternative high-frequency data from Google Trends provide a valuable proxy for real-time economic conditions. While there is a concern of unavailability of Google Trends data in these countries due to lack of internet access, politically motivated shutdowns, power outages, and failing infrastructure, the time period selected for this study largely mitigates these limitations; such disruptions were less pronounced during that interval.

Figure 1

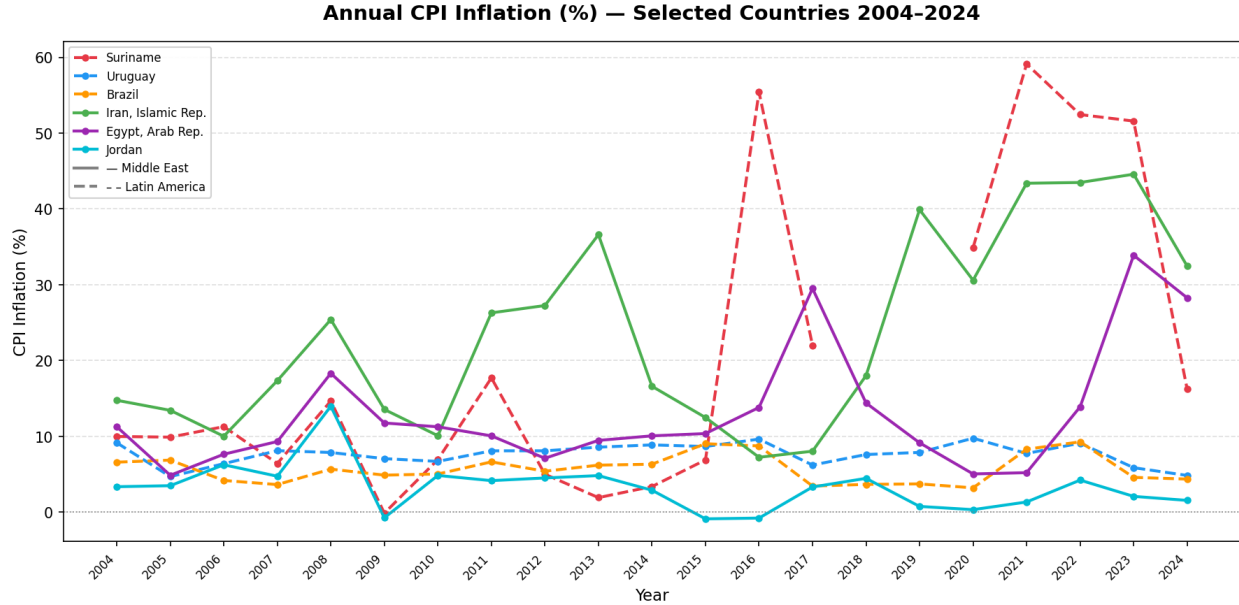


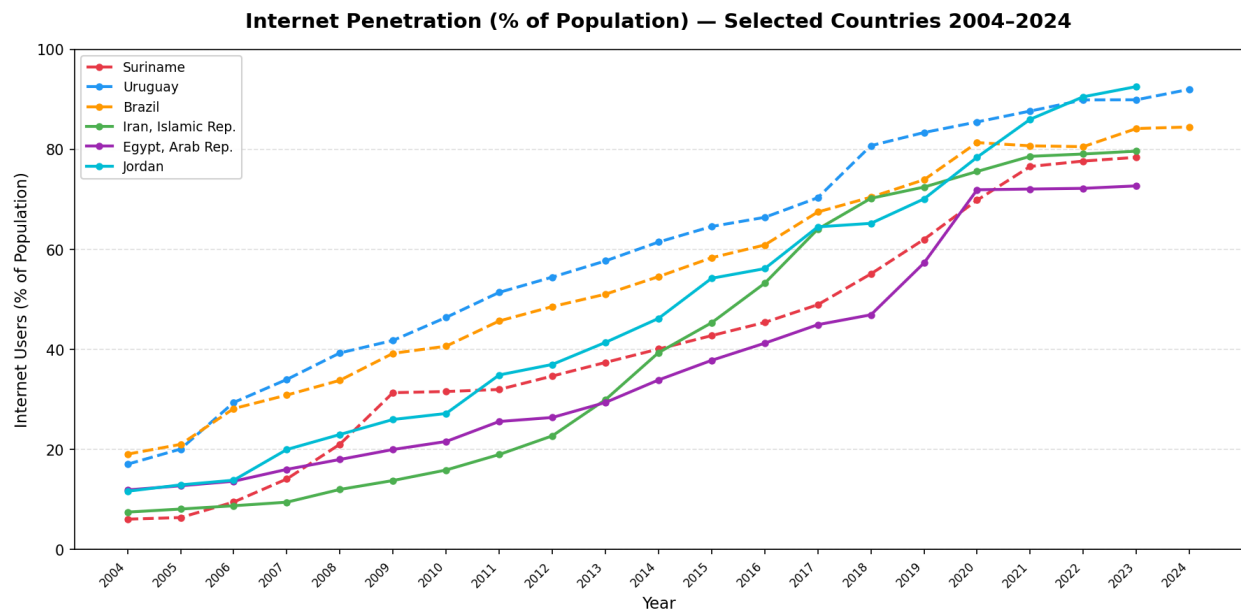
Figure 1 shows Annual CPI Inflation in percentage terms for our selected countries from 2004 to 2024. The three highest inflation rates over the time period are Suriname, Iran, and Egypt. Suriname peaks at almost 60% inflation in 2021, Iran at about 44% in 2023, and Egypt at about 34% in 2023. The three lowest inflation rates in Figure 1 are Jordan with negative inflation in 2009, 2015, and 2016, Brazil generally hovering between 0% and 10% inflation, and Suriname again with its dip at 0% in 2009 and near 2% in 2013. Suriname appears to be the most unpredictable and has the widest range of values given that its inflation minimum is about 0% in 2009 and then close to 60% in 2021. Even with reliable and reputable CPI data from the World Bank, Suriname still has missing data in 2018 and 2019. Missing observations for Suriname in 2018–2019 likely reflect data reporting limitations associated with institutional constraints and ongoing improvements in statistical systems. During this period, Suriname was transitioning to enhanced data dissemination standards under IMF guidance, while simultaneously undergoing post-crisis economic adjustment; this may have contributed to gaps or inconsistencies in reported macroeconomic data. Uruguay and Brazil do not appear to shift drastically over the time horizon and seem to have a lower variance compared to other countries.

B Country Selection Process

The data used in this study was obtained from the World Bank’s Open Data repository, a publicly accessible database providing comprehensive socioeconomic indicators across countries

and regions. The initial geographic scope of this study targeted two primary regions: Latin America, Central America, and the Middle East. To operationalize this selection, we employed the World Bank’s regional classification codes, specifically the MNA (Middle East and North Africa) code for the Middle Eastern region and the LCN (Latin America and the Caribbean) code for the Latin American and Central American region. It is important to note that both classification codes encompass broader geographic areas than those strictly relevant to the present analysis. The MNA code, for instance, includes both the Middle East and North Africa, while the LCN code extends beyond Latin America to also incorporate Central America and the Caribbean. To ensure geographic precision and maintain analytical focus, countries falling outside the defined regional scope were systematically excluded from the sample.

Figure 2



Specifically, from the LCN classification, the following Caribbean and island territories were excluded: Aruba, Antigua and Barbuda, the Bahamas, Barbados, the British Virgin Islands, the Cayman Islands, Curacao, Cuba, Dominica, the Dominican Republic, Grenada, Guadeloupe, Haiti, Jamaica, Puerto Rico, St. Kitts and Nevis, St. Lucia, St. Martin, St. Vincent and the Grenadines, Sint Maarten, Trinidad and Tobago, the Turks and Caicos Islands, and the U.S. Virgin Islands. Similarly, from the MNA classification, the following North African and peripheral territories were excluded: Algeria, Morocco, Tunisia, Libya, Djibouti, Malta, Sudan, South Sudan, Mauritania, and West Bank and Gaza.

This targeted exclusion process allowed for a more refined and theoretically consistent country

sample, ensuring that only countries within the intended regional boundaries were retained for analysis.

C Country Selection Filter

Following the delineation of regional boundaries outlined in the country selection process, a systematic multi-stage filtering procedure was applied to identify the most analytically appropriate countries for inclusion in the final sample. The selection process was governed by two quantitative eligibility filters: a ranking criterion and a fallback mechanism. The first filter assessed the availability and reliability of annual Consumer Price Index (CPI) data for each candidate country over the study period of 2004 to 2024, using the World Bank indicator FP.CPI.TOTL.ZG (Consumer Price Index, annual percentage change). To be retained in the sample, a country was required to have CPI observations for a minimum of 80% of the years within the study window, permitting a maximum of four missing observations out of a possible 21 years. This threshold was established to exclude countries with substantial data gaps attributable to prolonged conflict, political instability, or institutional collapse. Countries such as Syria, Yemen, and Libya, for instance, exhibit severe discontinuities in macroeconomic reporting that would render any forecasting model based on their data unreliable and methodologically inconsistent.

The second filter evaluated the adequacy of internet infrastructure as a precondition for generating a meaningful Google Search Volume Index (SVI) signal. Specifically, a country was required to maintain an average internet penetration rate exceeding 20% of the population over the 2004 to 2024 period, as measured by the World Bank indicator IT.NET.USER.ZS (Internet users as a percentage of the population). This criterion is grounded in the fundamental data-generating mechanism underlying Google Trends: search volume indices are only statistically reliable when a sufficiently large share of the population actively uses the internet. In countries where internet adoption remains critically low, SVI signals tend to be sparse, noisy, and unrepresentative of broader population behavior, properties that substantially undermine their utility as proxies for inflation-hedging sentiment. Figure 2 shows the percentage of population in each country that uses the internet over the research timeline.

Countries satisfying both eligibility filters were subsequently ranked in descending order of their average annual CPI inflation rate over the 2004 to 2024 period, and the top three countries per region were included in the final analytical sample. This ranking criterion reflects a deliberate methodological preference for countries exhibiting higher and more volatile inflation environments, as such contexts provide richer variation in the dependent variable and stronger. This leads to more discernible signals for the forecasting models to exploit from Google Trends data, since a larger base of internet users generates a higher volume of inflation-related search activity, producing a

clearer and more representative signal of what the public is searching. Internet penetration, in this context, is measured as the share of users who have maintained an internet connection for at least three consecutive months. If there was a case after applying the two filters where only three countries could pass the selection criteria, a fallback procedure was applied whereby the pipeline automatically supplemented the qualifying set with the next best-ranked candidates by average CPI inflation, subject to the condition that countries with no available data whatsoever were categorically excluded. This contingency mechanism preserves the structural symmetry of the cross-regional design while maintaining a minimum standard of data quality.

D Google Trend Keyword Selection

To create a standardized keyword selection strategy across all countries, we searched for the same keywords across each country. We created four different channels of keywords to capture consumer sentiment and inflation expectations accurately: Consumer Price Index (CPI), Currency and Investment, Labor Market, and Financial Stress. Following the keyword selection method from Bikourne et al. (2025), we searched for the words in their respective local language and in English. The countries we selected for our study speak the following languages: English, Dutch, Spanish, Portuguese, Persian, and Arabic. We used DeepL to translate English words into the local language. The Google Trends publicly available data allows for only five keywords to be downloaded at a time for any given country. Thus, we downloaded each keyword in .csv files and merged them by date. Refer to section C of the Appendix for a full dictionary list of the keywords by channel. In the CPI channel, we selected 29 words from the United Nations 2018 Classification of Individual Consumption According to Purpose (COICOP). The COICOP is helpful because it anchors our analysis with a consistent and universal framework to follow for CPI keyword selection across countries, rather than relying on subjectivity in our choices.

For the Currency and Investment channel, we decided to focus on two key ideas: exchange rate anxiety and capital flow. To determine how we would select the keywords in this channel for exchange rate anxiety, we referenced chapters 13, 14, and 15 in Krugman et al.'s textbook "International Economics Theory & Policy." To obtain further research to support our keyword selection for capital flows, we follow the IMF's institutional view of capital flow, specifically a 2012 policy paper that gave advice to countries about how to better manage capital flows. We also used a few keywords in the Currency and Investment, Labor Market, and Financial Stress channels from Castelnuovo and Tran (2017) given their conclusion of improved unemployment movement predictions, such as *bankruptcy*, *financial crisis*, *unemployment benefit*, and *inflation*.

Labor Market channel keywords were selected using a few different sources. The word *jobs* was helpful in D'Amuri and Marcucci's (2017) research using Google Trends keyword data to

forecast U.S. unemployment because it proved “to be the most popular among different job-search-related keywords.” Chadwick and Sengül (2015) used twenty different keywords to help nowcast the unemployment rate in Turkey, of which we also used the words *unemployment*, *CV*, and *resume*.

V Model Specification

Specification 1:

$$\text{Inflation} = \alpha + \beta_1(\text{Inflation}) + \varepsilon$$

Specification 2:

$$\text{Inflation} = \alpha + \beta_1(\text{Google Trends}) + \varepsilon$$

Specification 3:

$$\text{Inflation} = \alpha + \beta_1(\text{CPI}) + \beta_2(\text{Unemployment Rate}) + \beta_3(\text{Exchange Rate in USD}) + \beta_4(\text{M2}) + \varepsilon$$

Specification 4:

$$\text{Inflation} = \alpha + \beta_1(\text{CPI}) + \beta_2(\text{Unemployment Rate}) + \beta_3(\text{Exchange Rate in USD}) + \beta_4(\text{M2}) + \beta_5(\text{Google Trends}) + \varepsilon$$

Table 2: Yearly Model Specifications

Specification	Model
Specification 1	Inflation ~ Inflation
Specification 2	Inflation ~ Macro variables
Specification 3	Inflation ~ Macro variables + Google Trends (Composite Index A)
Specification 4	Inflation ~ Google Trends (Composite Index A)
Specification 5	Inflation ~ Composite Index B
Specification 6	Inflation ~ Composite Index C
Specification 7	Inflation ~ Financial Stress Channel
Specification 8	Inflation ~ Labor Market Channel
Specification 9	Inflation ~ Currency and Investment Channel
Specification 10	Inflation ~ CPI Channel

A Results

We estimated two models, ARIMA and ARIMAX, across ten specifications (Table 2) using annual data. The baseline specification is an autoregressive integrated moving average (ARIMA) model that forecasts inflation using the lagged values of inflation. The remaining nine specifications augment the baseline with exogenous variables and are evaluated relative to it in terms of forecast accuracy. The sample spans 2004–2024, with 2004–2017 used for training and 2018–2024 reserved

Table 3 : Monthly Model Specifications

Specification	Model
Specification 1	Inflation ~ Inflation
Specification 2	Inflation ~ Labor Market Channel (corr)
Specification 3	Inflation ~ Financial Stress Channel (corr)
Specification 4	Inflation ~ Currency and Investment Channel (corr)
Specification 5	Inflation ~ CPI Channel (corr)
Specification 6	Inflation ~ Labor Market Channel (equal)
Specification 7	Inflation ~ Financial Stress Channel (equal)
Specification 8	Inflation ~ Currency and Investment Channel (equal)
Specification 9	Inflation ~ CPI Channel (equal)
Specification 10	Inflation ~ Google Trends (Composite Index A)
Specification 11	Inflation ~ Google Trends (Composite Index B)
Specification 12	Inflation ~ Google Trends (Composite Index C)

for out-of-sample testing. Baseline forecast results for all six countries are presented in Figures 1–6 in the Appendix. Model performance is assessed using the Root Mean Squared Error (RMSE). Least RMSE means that a model has the smallest average deviation between its predicted values and the actual observed values, with larger errors penalized more heavily. In forecasting contexts, RMSE is a standard measure of accuracy, so a lower RMSE indicates more precise predictions. Therefore, the “best model” in this framework is the one that minimizes RMSE on the test dataset, implying it provides the most reliable out-of-sample forecasts among the competing specifications.

Across the South American countries, macroeconomic variable specifications yielded the lowest RMSE in Uruguay (RMSE: 1.00 unit) and Brazil (RMSE: 1.18 units), while in Suriname the Labor Market channel produced the best performance with an RMSE of 21.51 units. Among these, Uruguay achieved the highest forecast accuracy overall under macroeconomic specification. Figure 3 illustrates the corresponding forecast for Uruguay.

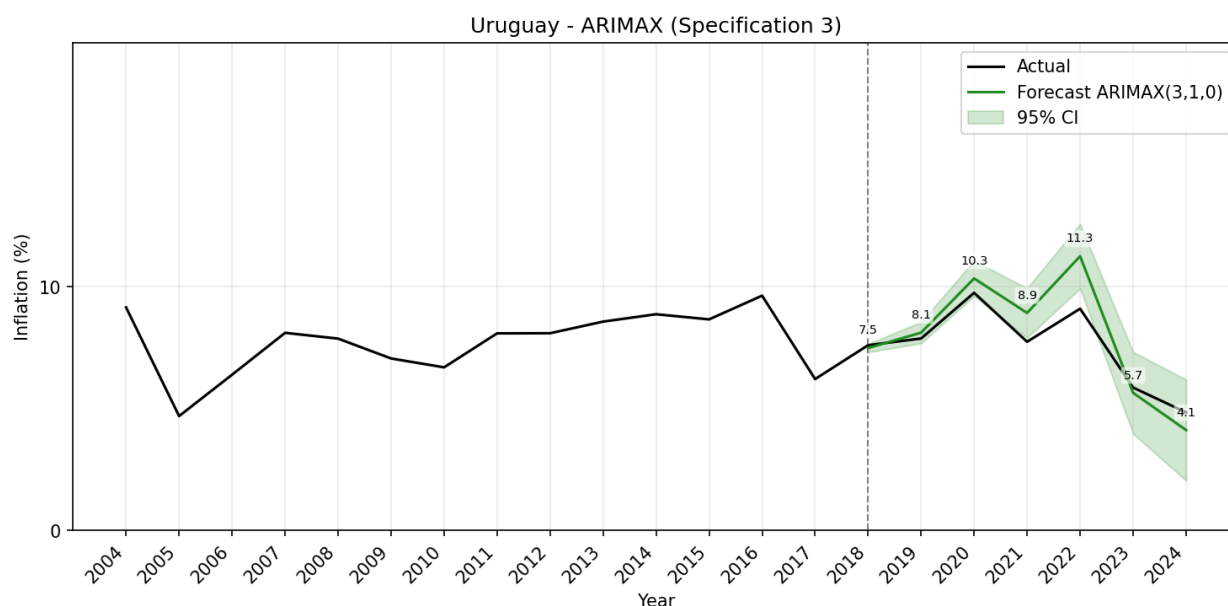
In Middle Eastern countries, the Financial Stress channel performed best in Iran, with an RMSE of 20.98 units, while the Labor Market channel generated the lowest RMSE in Egypt (RMSE: 10.75 units) and Jordan (RMSE: 0.71 units). Among these, Jordan exhibited the highest forecast accuracy, with the Labor Market channel outperforming other specifications. Figure 4 presents the forecast for Jordan under this specification. To assess model adequacy, Ljung–Box tests were conducted to evaluate residual autocorrelation, with results reported in Table 4. To further quantify model improvement, we computed the percentage reduction in RMSE relative to the baseline model. The gains range from 5% to 56%, as reported in Table 5. We also observed that confidence intervals widen with forecast horizon as multi-step errors compound through AR and MA coefficients. The baseline width reflects training-period residual variance, while the rate of widening is governed by

model order.

The cross-country results indicate that forecasting performance is closely linked to the dominant inflation transmission mechanisms within each economy. In Egypt, Jordan, and Suriname, the Labor Market channel consistently produced the lowest RMSE, suggesting that labor-related search behavior captures inflationary pressures more effectively than traditional macroeconomic indicators in these contexts. In Suriname, where structural breaks, hyperinflation, and institutional weaknesses undermine the reliability of official data, labor-related search intensity likely proxies real-time household responses to declining purchasing power and IMF-led adjustment. In Egypt, repeated exchange-rate devaluations, subsidy reforms, and food price shocks have transmitted inflation through labor market insecurity and expectations, making labor-related searches a timely indicator of underlying price dynamics. In Jordan, where the exchange rate peg constrains inflation variability, persistent unemployment and labor market fragility appear to provide a more informative signal than relatively stable macroeconomic variables.

Similarly, the strong performance of the Financial Stress channel in Iran highlights the importance of demand-side pressures and household financial constraints in a sanctions-driven economy. Search behavior related to financial stress, subsidies, and credit conditions may capture underlying inflation dynamics that are not fully reflected in official macroeconomic data, particularly in the presence of multiple exchange rates and data limitations.

Figure 3



By contrast, macroeconomic variables produced the lowest forecast error in Brazil and Uruguay,

suggesting that inflation in these economies remained more closely linked to structured macroeconomic fundamentals, including exchange-rate pass-through, commodity price shocks, and monetary policy responses. In Brazil, where RMSE declined from 2.16 units in the baseline specification to 1.18 units in the best-performing model, macroeconomic variables likely captured post-pandemic inflation dynamics associated with energy prices, food shocks, and monetary tightening more effectively than search-based indicators. In Uruguay, where inflation is characterized by wage indexation, external-price pass-through, and strong links to regional economic conditions, macroeconomic variables appear to have aligned more closely with the underlying inflation process, consistent with the reduction in RMSE from 1.63 units to 1.00 units.

Figure 4

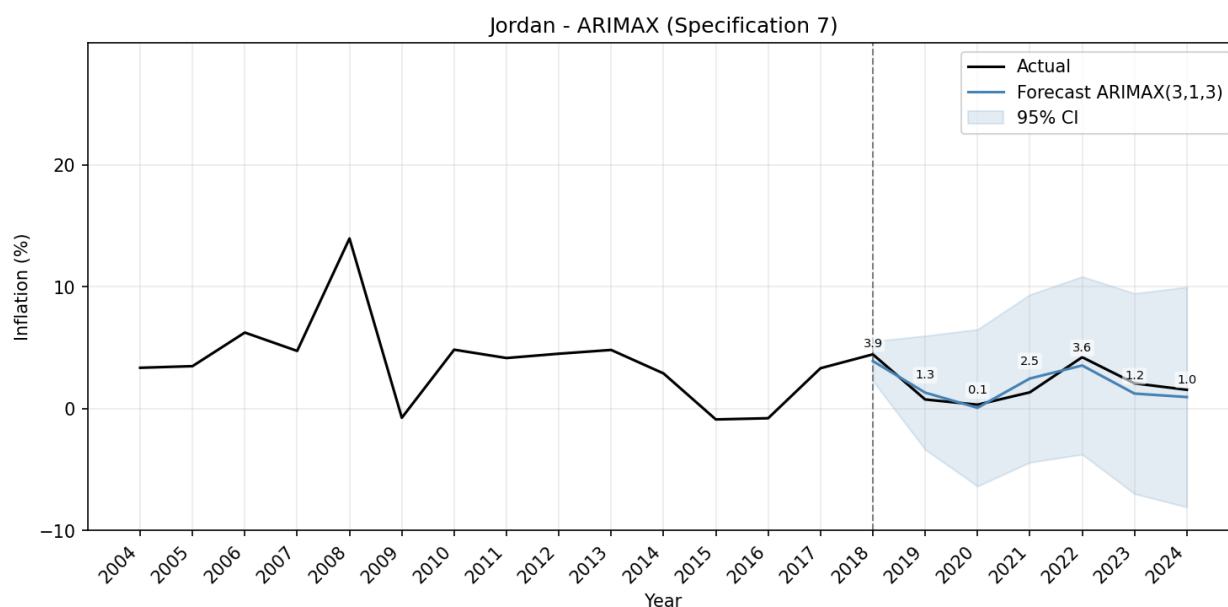


Table 4 : Best models - Yearly Frequency

	Suriname	Uruguay	Brazil	Iran	Egypt	Jordan
Specification	ARIMAX	ARIMAX	ARIMAX	ARIMAX	ARIMAX	ARIMAX
Exogenous variable	Labour Market	Macro Variables	Macro Variables	Financial Stress	Labour Market	Labour Market
Order (p, d, q)	(0, 1, 2)	(3, 1, 0)	(2, 1, 2)	(0, 1, 3)	(0, 1, 3)	(3, 1, 3)
Exogenous lag	2	2	2	2	2	2
RMSE	21.51	1.00	1.18	20.98	10.75	0.71
Ljung-Box test p -value	0.9370	0.0677	0.0811	0.3131	0.7224	0.5536

Note: Best specification selected by lowest RMSE across all 10 specifications. Ljung-Box p -value > 0.05 indicates white noise residuals.

Table 5: Best Specification vs. Baseline by Country (Yearly Multi-Step Forecasts)

Country	Best Specification	Base RMSE	Best RMSE	Change (%)
Brazil	Inflation ~ Macro Variables	2.16	1.18	−45.4%
Egypt	Inflation ~ Labour Market Channel	17.31	10.75	−37.9%
Iran	Inflation ~ Financial Stress Channel	29.30	20.98	−28.4%
Jordan	Inflation ~ Labour Market Channel	1.62	0.71	−56.2%
Suriname	Inflation ~ Labour Market Channel	22.87	21.51	−5.9%
Uruguay	Inflation ~ Macro Variables	1.63	1.00	−38.7%

¹. Base model is the autoregressive baseline (Inflation ~ Inflation).

². Best Specification selected by lowest out-of-sample RMSE across all ten specifications.

³. Negative values indicate RMSE *reduction* — the best specification outperforms the baseline.

Similarly, we ran two models (ARIMA and ARIMAX) on twelve specifications (Table 3) for monthly frequency. The monthly macroeconomic variables data was not available for the selected countries. We used the four channels of the Google Trends SVI and the three composite indexes constructed from the four channels (Composite Index A, Composite Index B, Composite Index C) to test our theory as to whether the Google Trends data improves forecast accuracy or not. We divided out monthly data ranging from 2004 to 2024 into test and training datasets where March 2023 to January 2024 were the test range. Figures (1-a) to (7-a) in the Appendix show the baseline model's forecast results for all seven countries. Among the twelve specifications, the CPI channel yielded the lowest RMSE in Uruguay, while the Financial Stress channel performed best in Argentina. In Brazil, Composite Index A emerged as the most accurate specification. In contrast, for Suriname, neither the individual channels nor the composite indices improved forecast performance; instead, the baseline model relying on lagged inflation produced the most accurate forecasts.

Out of the four South American countries, Uruguay has the least rolling and multi-step RMSE. The forecast produced by the CPI channel in Uruguay can be seen in Figure 5. The Financial Stress channel in Argentina has the highest rolling and multi-step RMSE. Figure 6 displays the forecast in Argentina.

Figure 5

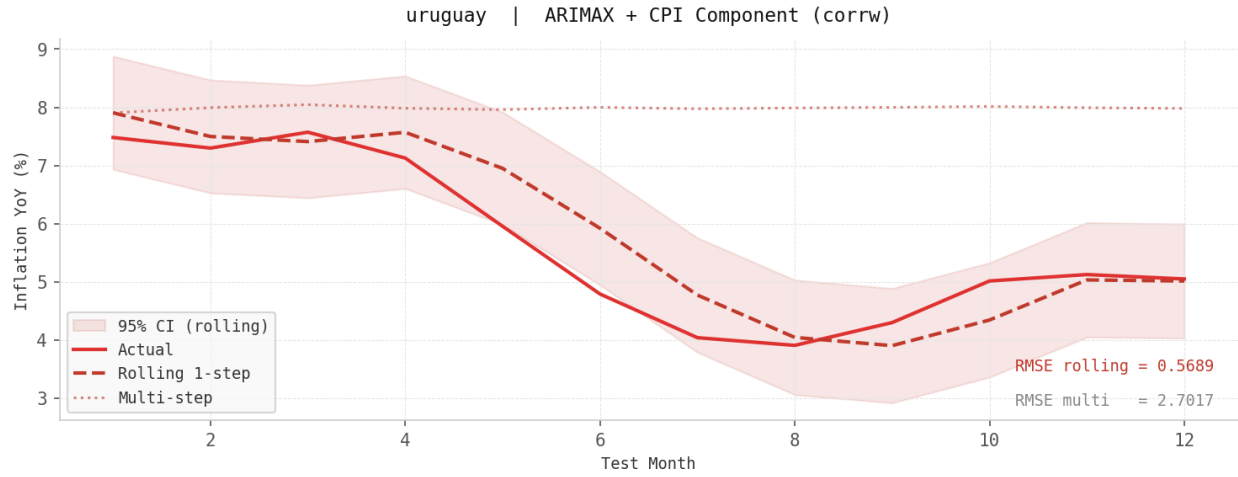
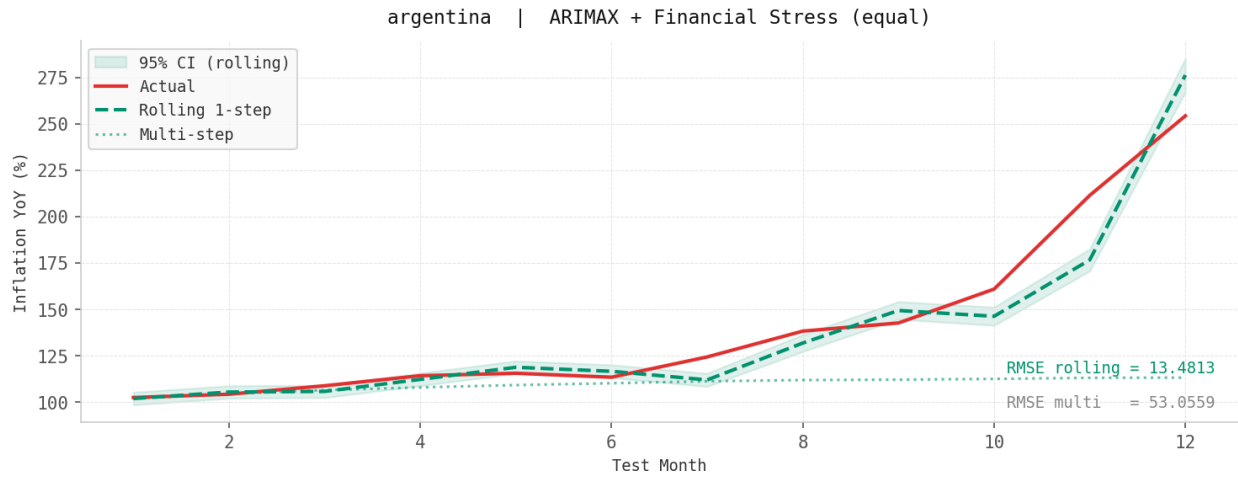
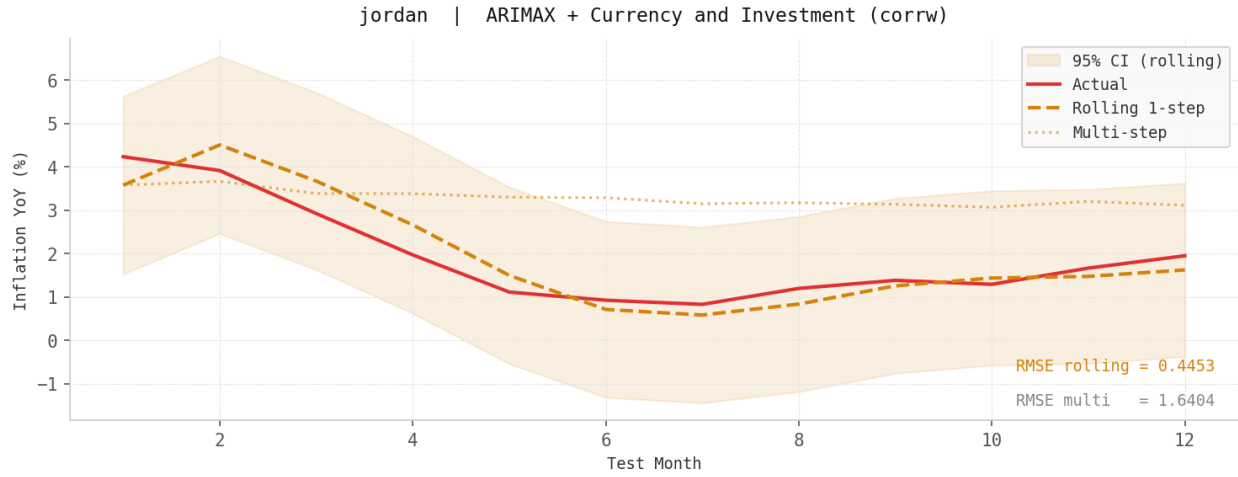


Figure 6



In Middle Eastern countries, the CPI channel produced the least RMSE in Egypt. Composite Index A in Iran and the Currency and Investment channel in Jordan produced accurate inflation forecast results. Out of all the Middle Eastern countries, Jordan has the least rolling and multi-step RMSE. The forecast produced by the Currency and Investment channel in Jordan can be seen in Figure 7. The rolling RMSE and the multi-step RMSE produced by each exogenous variable in each country and the results from the Ljung-Box test can be seen in Table 6.

Figure 7**Table 6 : Best models - Monthly Frequency**

	Argentina	Brazil	Egypt	Iran	Jordan	Suriname	Uruguay
Specification	ARIMAX	ARIMAX	ARIMAX	ARIMAX	ARIMAX	ARIMA	ARIMAX
Exogenous variable	Financial Stress	Composite A	CPI Component	Composite A	Currency & Inv.	—	CPI Component
Weighting	equal weights	equal weights	correlation-weighted	equal weights	correlation-weighted	—	correlation-weighted
Order (p, d, q)	(1, 1, 0)	(2, 0, 0)	(3, 0, 1)	(2, 1, 2)	(2, 1, 0)	(1, 1, 0)	(3, 1, 0)
Exogenous lag	1	1	1	4	2	—	1
RMSE (rolling)	13.4813	0.5120	2.0752	3.8999	0.4453	4.7163	0.5689
RMSE (multi-step)	53.0559	1.1786	6.2172	9.8819	1.6404	12.9665	2.7017
Ljung-Box test p -value	0.6299	0.0063	0.9390	0.9305	0.0549	0.5853	0.9261

Note: Best specification selected by lowest RMSE across all 12 specifications. Ljung-Box p -value > 0.05 indicates white noise residuals.

The algorithm runs through each country's inflation data with auto ARIMA (from the Python *pmdarima* package) to determine the best-fit inflation ARIMA model based on AIC. The auto ARIMA function runs multiple permutations and combinations through the range provided to select the best order (p, d, q). The best order is determined by least AIC. Upon running this on each country, we used the Granger Causality test to test between lag 1 to 6 for the exogenous variable Google Trends SVI data. It flags the statistically significant ones and simulates a 12 step-ahead forecast and records RMSE of the forecast and compares all significant models to determine the best forecast model for that country. To isolate the effect of a single channel and avoid potential multicollinearity issues, specifications 2 to 9 only include the channel, while specifications 10 to 12 include the composite index, which includes all signals from four channels. Equal weight channels' score takes a simple average of all the individual keywords in that channel, while correlation weighted channels' score weighs them based on the pearson correlation of a given keyword against inflation. The idea is to select keywords that are more likely to contain relevant information,

increase signal, and reduce noise. The same logic applies to the survival rate weighted composite index. If a channel has more keywords that pass the filter, it could contain more useful information for the forecast and thus it is given more weight. All specifications include all versions of the channel scores and composite index to test for robustness. This allows us to test which mechanism provides the most useful information for forecasting inflation.

Two things to note in the summary table are the large RMSE for Argentina and the best model for Brazil rejecting the Ljung-Box test, suggesting its residuals are not white noise. These are limitations of our models. Brazil underwent a central bank policy change in mid-2023, which falls in the middle of our testing window. This shift in monetary policy created a structural change in the data generating process of inflation that cannot be captured by a model trained on data from previous years, thus leaving temporal structures in the residuals. Argentina, on the other hand, has experienced volatile inflation since early 2023 that persisted through 2024, covering the entire training window. These two countries serve as examples of how our approach can face limitations.

Table 7: Best Specification vs. Baseline by Country (Monthly Rolling Forecasts)

Country	Best Specification	Base RMSE	Best RMSE	Change (%)
Argentina	Inflation ~ Financial Stress Channel (equal)	14.2239	13.4813	-5.22%
Brazil	Inflation ~ Google Trends (Composite Index A)	0.5168	0.5120	-0.93%
Egypt	Inflation ~ CPI Channel (corr _w)	2.2708	2.0752	-8.61%
Iran	Inflation ~ Google Trends (Composite Index A)	4.0867	3.8999	-4.57%
Jordan	Inflation ~ Currency and Investment Channel (corr _w)	0.4733	0.4453	-5.92%
Suriname	Inflation ~ Inflation (baseline)	4.7163	4.7163	0.00%
Uruguay	Inflation ~ CPI Channel (corr _w)	0.5771	0.5689	-1.42%

¹. Base model is the autoregressive baseline (Inflation ~ Inflation).

². Change is percentage difference in rolling RMSE. Suriname showed no improvement.

³. (*corr_w*) = correlation-weighted; (*equal*) = equal-weighted.

To evaluate how well each best model performed when compared to the base model, the forecast error was calculated. The percentage change in the RMSE of the baseline specification and the best specification for each country ranged between 0% to 8.61%. Table 7 shows the results of the percentage difference between the base model RMSE and and best model RMSE for monthly specifications. Suriname's percent change of 0% comes from its best monthly model not using exogenous variables or weighting; a simple ARIMA(1,1,0) rendered the lowest rolling RMSE. The highest percentage change in the base RMSE and the best RMSE was for Egypt at 8.61%. The top two specifications appeared twice in Table 7: Inflation ~ Google Trends (Composite Index A) for Brazil and Iran, and Inflation ~ CPI Channel (corr_w) for Egypt and Uruguay.

Overall, these findings suggest that Google Trends-based channels contributed most where inflation dynamics were shaped by household distress and expectation formation, while traditional macroeconomic predictors performed better where inflation was driven by more conventional and

institutionally mediated channels. This cross-country heterogeneity reinforces the importance of aligning predictor selection with country-specific inflation regimes rather than assuming a single forecasting specification performs uniformly across different macroeconomic contexts.

VI Limitations

Our research is not without its limitations. A fundamental challenge inherent to the use of alternative data sources such as internet search activity is that while they may correlate with economic outcomes, they cannot establish causality. Search volume reflects what consumers are thinking about or reacting to, but it does not reveal the underlying mechanisms driving those behaviors. It also cannot directly mean that shifts in search activity are the cause of inflationary pressures. Thus, any predictive relationship identified in this study should be interpreted as associative rather than structural. While this study demonstrates the potential of Google Trends data for inflation forecasting, several limitations warrant acknowledgment.

1. First, keyword selection poses a fundamental challenge. Identifying terms that accurately reflect consumer sentiment in developing economies is inherently difficult. For example, certain economically relevant topics in the 2010s onwards such as cryptocurrency did not exist in the mid-2000s, creating gaps in historical coverage.

2. The availability of limited time-series observations across several countries increases the risk of overfitting, particularly in the more parameter-intensive non-classical models such as RNN and LSTM variants.

3. The composite index construction mechanism remains a "black box" given the five word search limit per query with the *pytrends* Python library. Every five word query's reference group is the highest number in the query, whereas knowing the reference group for all keywords across channel would be more ideal. When we searched for a keyword, it returned a search volume. We do not know the gross amount of people searching for the terms. An SVI of 50 does not tell us how many people searched for a given term. While the channel-based approach covering labor market, financial stress, currency and investment, and CPI components provides structure, it cannot fully capture the true intent behind search queries. We do not know the characteristics of the person searching for the term. For example, we cannot determine whether a search for *oil price* is simply out of curiosity or because a user's business depends on it. Each word serves as a proxy of consumer expectation. A single keyword may reflect diverse motivations across different user populations.

4. The reliance on annual CPI data for several countries constrains model choice, reliability, and usefulness. Higher frequency monthly, weekly, or even daily macroeconomic data would be the most ideal for capturing short-term inflation dynamics.

5. We cannot account for users who connect to a Virtual Private Network (VPN) and the degree

of interconnectedness among countries. An individual may search for a term while physically located in the United States, but may be using a VPN to mask their true IP address to Brazil. This may bias our results. For example, the interconnectedness among Brazil, Argentina, and Uruguay may mean that the searches for a particular word in Brazil might help forecasts in Uruguay. We believe this is an important step for future research.

6. Finally, the geographic scope of the Google Trends index is national, which may mask meaningful sub-national or regional variation in consumer behavior. A more localized trend index could yield sharper and more actionable insights.

VII Conclusion

This study evaluated the predictive power of Google Trends Search Volume Index (SVI) data for short-term inflation forecasting across a diverse set of developing economies using classical modeling approaches. In the classical monthly models, ARIMAX specifications consistently delivered strong performance across all seven countries: Argentina, Brazil, Egypt, Iran, Jordan, Suriname, and Uruguay. Rolling RMSE values ranging from 0.45 units in Jordan to 13.48 units in Argentina, and incorporating composite Google Trends indices and CPI-component channels improved forecast accuracy in several cases. Including the Google Trends index reduced rolling RMSE on average about 4 percent compared to simple ARIMA models. At the yearly frequency, the classical ARIMAX models augmented with macroeconomic exogenous variables, including labor market indicators, macroeconomic variables, and financial stress channels produced well-specified forecasts, with RMSE values as low as 0.71 units for Jordan and 1.00 unit for Uruguay while reducing the average RMSE by about 30 percent compared to simple ARIMA models. The Ljung-Box test's p-values exceeded 0.05 for most countries, confirming white-noise residuals. These results suggest that Google Trends data contributes more to forecast accuracy at lower frequency levels, with annual models showing a substantially larger reduction in RMSE compared to monthly models. Taken together, these findings confirm that integrating Google Trends SVI into classical forecasting frameworks can enhance inflation prediction accuracy in data-scarce environments, with the degree of improvement varying by country context, forecast length, and model architecture. These results indicate that search-based data can be used as a complement to macroeconomic data instead of substitution. Future work can explore country-specific keywords which fully capture local language dialects to further improve forecast accuracy.

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Appendix

A Descriptive Statistics by Country

Annual Descriptive Statistics — Brazil (2004–2024)

Variable	N	Mean	Median	SD	Min	Max
CPI Index	21	135.66	126.93	46.72	74.43	223.22
Inflation (WB, %)	21	5.71	5.40	1.92	3.21	9.28
Exchange Rate (LCU/USD)	21	3.19	2.93	1.34	1.67	5.39
Unemployment Rate (%)	21	9.64	9.28	2.20	6.75	13.70
Money Supply M2 (% of GDP)	21	85.41	81.91	17.34	56.33	116.39
Internet Penetration (% pop.)	21	54.99	54.55	21.58	19.07	84.46
Oil Price (USD)	21	74.57	71.61	23.01	38.23	112.01

Annual Descriptive Statistics — Egypt (2004–2024)

Variable	N	Mean	Median	SD	Min	Max
CPI Index	21	201.29	142.05	152.58	55.08	623.83
Inflation (WB, %)	21	13.08	10.37	8.06	4.87	33.88
Exchange Rate (LCU/USD)	21	12.50	7.08	10.10	5.43	45.30
Unemployment Rate (%)	21	9.92	9.86	2.27	6.55	13.15
Money Supply M2 (% of GDP)	21	84.91	83.70	9.09	69.72	98.14
Internet Penetration (% pop.)	20	37.32	31.65	21.70	11.92	72.69
Oil Price (USD)	21	74.57	71.61	23.01	38.23	112.01

Annual Descriptive Statistics — Iran (2004–2024)

Variable	N	Mean	Median	SD	Min	Max
CPI Index	21	539.34	256.00	749.04	43.56	2834.85
Inflation (WB, %)	21	23.41	18.01	12.68	7.25	44.58
Exchange Rate (LCU/USD)	21	24702.00	25941.66	14543.85	8613.99	42000.00
Unemployment Rate (%)	21	10.96	10.77	1.49	8.13	13.68
Money Supply M2 (% of GDP)	21	63.40	60.96	15.84	38.76	79.14
Internet Penetration (% pop.)	20	40.23	34.65	28.53	7.49	79.63
Oil Price (USD)	21	74.57	71.61	23.01	38.23	112.01

Annual Descriptive Statistics — Jordan (2004–2024)

Variable	N	Mean	Median	SD	Min	Max
CPI Index	21	110.10	115.49	19.81	73.20	138.01
Inflation (WB, %)	21	3.27	3.36	3.22	-0.88	13.97
Exchange Rate (LCU/USD)	21	0.71	0.71	0.00	0.71	0.71
Unemployment Rate (%)	21	15.06	14.63	2.57	11.90	19.84
Money Supply M2 (% of GDP)	21	119.17	115.74	10.25	103.79	138.88
Internet Penetration (% pop.)	20	47.59	43.80	26.85	11.66	92.53
Oil Price (USD)	21	74.57	71.61	23.01	38.23	112.01

Annual Descriptive Statistics — Suriname (2004–2024)

Variable	N	Mean	Median	SD	Min	Max
CPI Index	21	352.74	130.25	463.70	62.74	1698.84
Inflation (WB, %)	19	20.30	11.28	19.92	-0.13	59.12
Exchange Rate (LCU/USD)	21	8.89	3.30	10.34	2.73	36.78
Unemployment Rate (%)	21	8.12	7.92	0.83	6.60	9.49
Money Supply M2 (% of GDP)	21	57.87	55.04	13.17	38.73	84.92
Internet Penetration (% pop.)	20	41.05	38.74	23.41	6.08	78.40
Oil Price (USD)	21	74.57	71.61	23.01	38.23	112.01

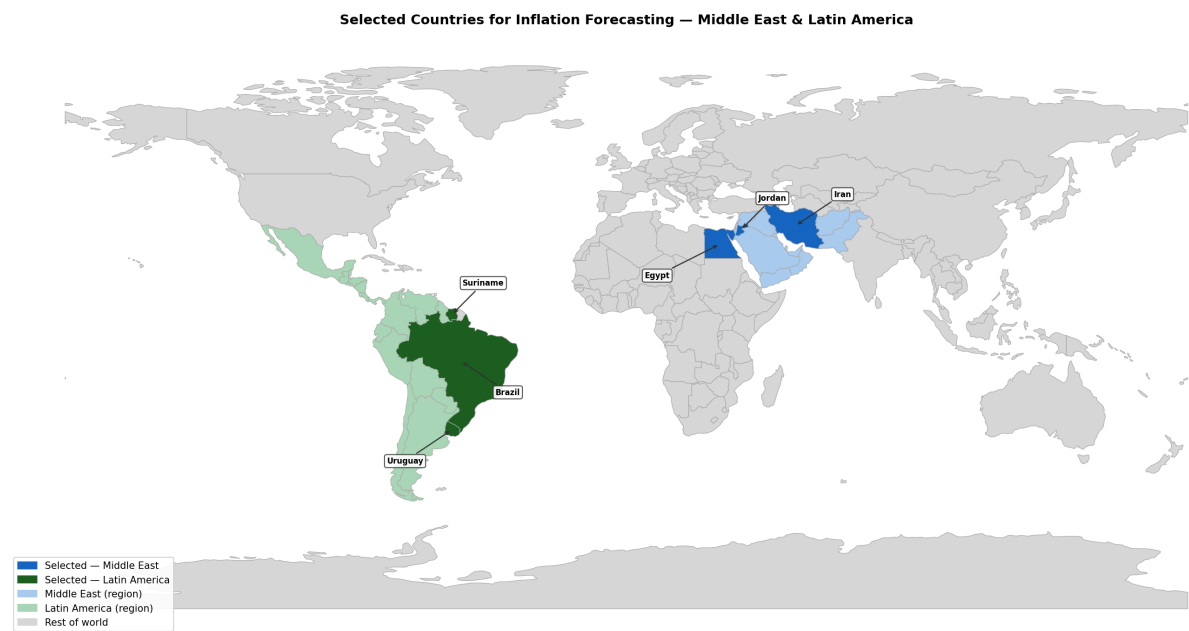
Annual Descriptive Statistics — Uruguay (2004–2024)

Variable	N	Mean	Median	SD	Min	Max
CPI Index	21	154.20	138.13	71.82	67.38	290.63
Inflation (WB, %)	21	7.67	7.88	1.44	4.70	9.76
Exchange Rate (LCU/USD)	21	28.84	27.33	8.17	19.31	43.55
Unemployment Rate (%)	21	8.50	8.03	1.80	6.31	12.98
Money Supply M2 (% of GDP)	21	46.72	45.28	6.01	37.82	59.13
Internet Penetration (% pop.)	21	60.16	61.46	23.95	17.06	91.99
Oil Price (USD)	21	74.57	71.61	23.01	38.23	112.01

Monthly CPI Inflation — All Countries (2004–2024)

Country	N	Mean	Median	SD	Min	Max
Argentina	97	1294.67	385.88	2030.66	100.00	7694.00
Brazil	252	98.03	91.84	33.05	52.12	164.74
Egypt	177	88.17	80.81	49.15	33.91	231.40
Iran	249	48.48	23.25	65.74	3.83	285.10
Jordan	227	91.24	93.43	13.10	62.64	111.17
Suriname	238	176.05	62.79	221.54	31.26	836.10
Uruguay	240	59.09	53.24	25.80	25.68	110.29

A Country Map



B Base Model

A Middle Eastern Countries

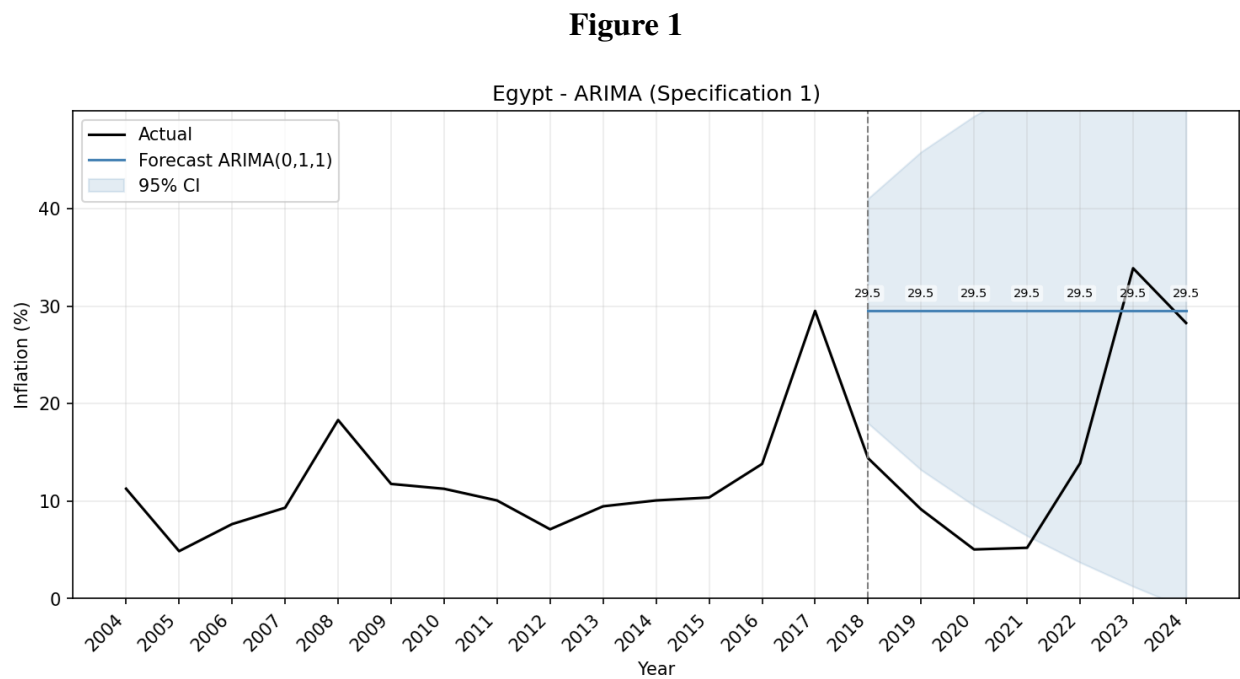


Figure 1-a

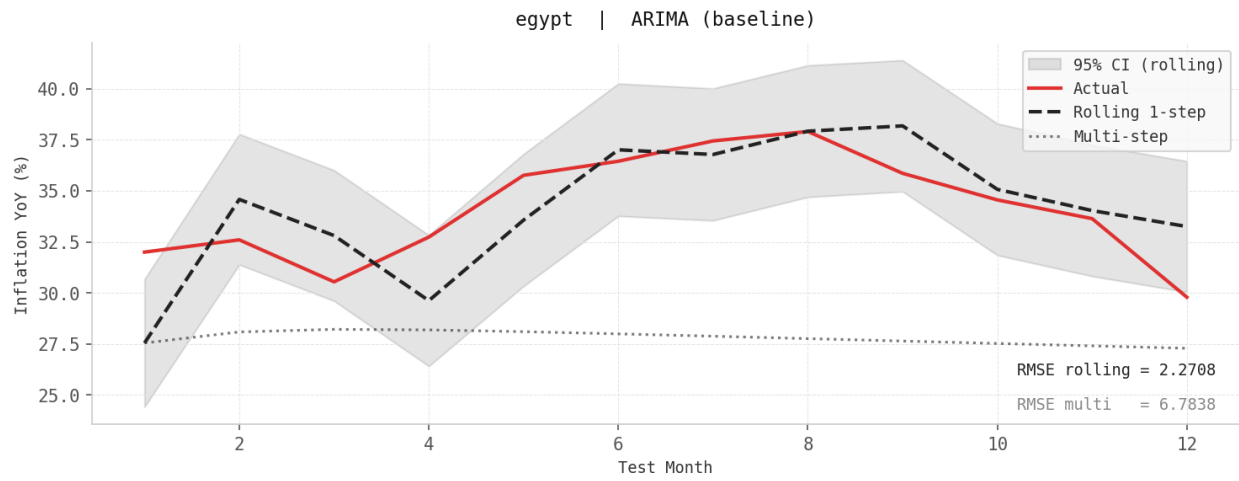


Figure 2

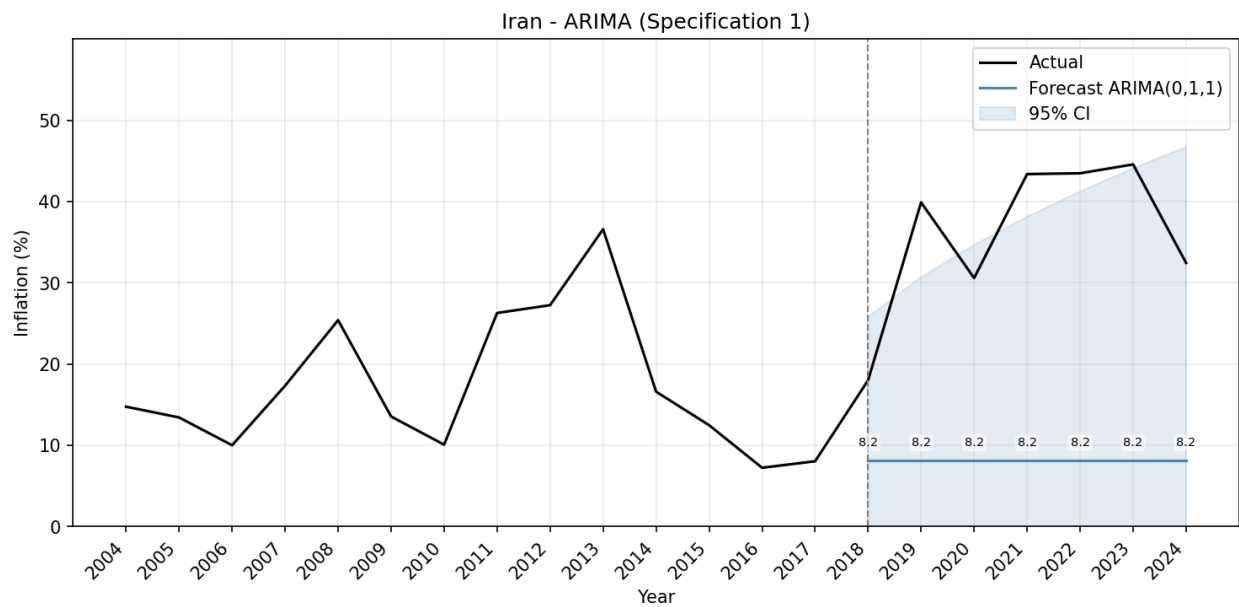


Figure 2-a

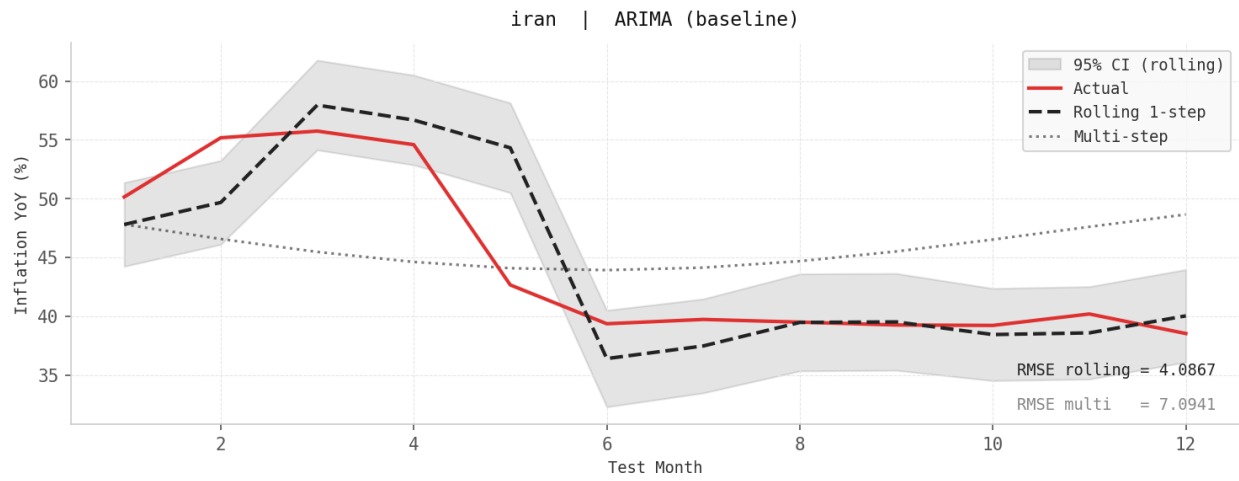


Figure 3

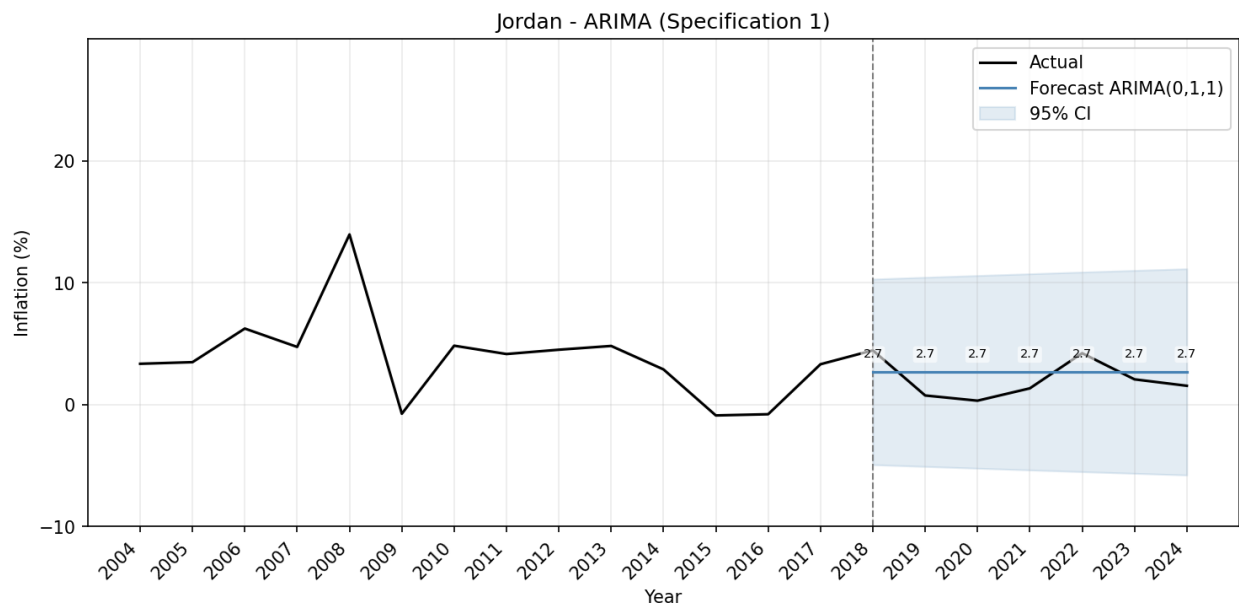
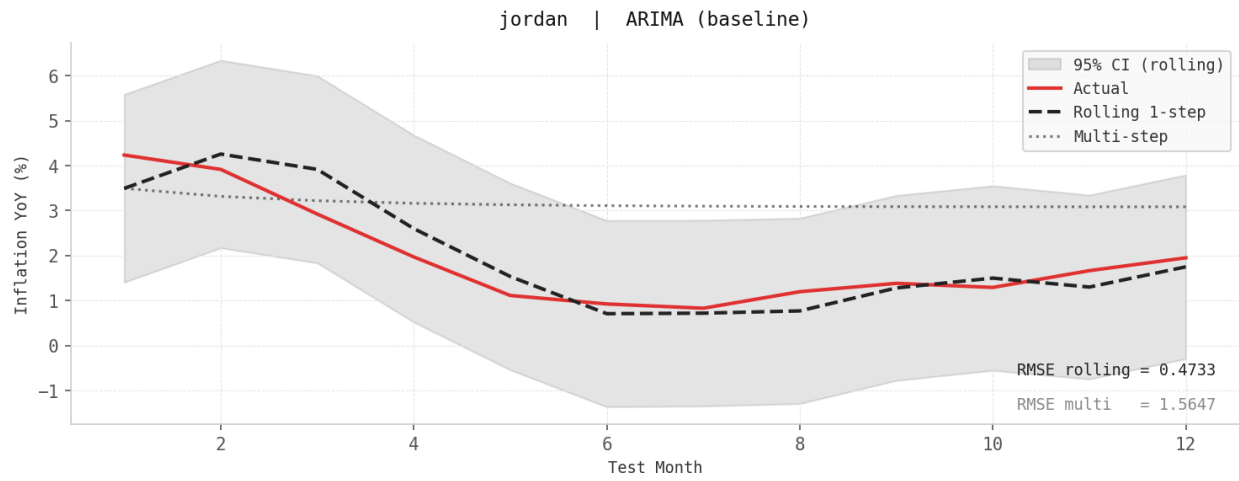


Figure 3-a



B South American Countries

Figure 4

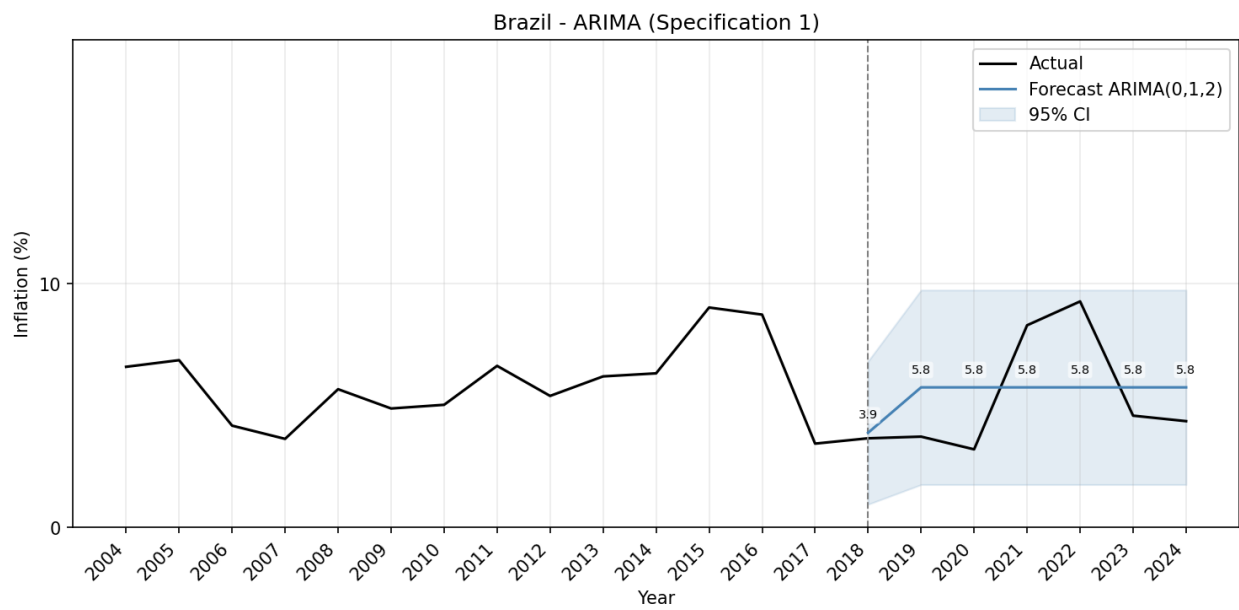


Figure 4-a

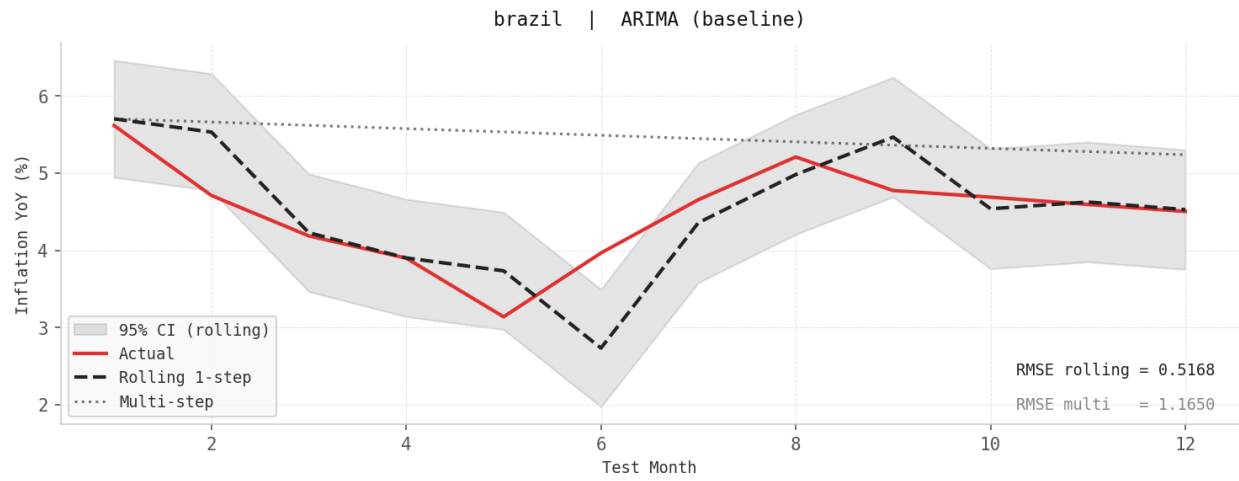


Figure 5

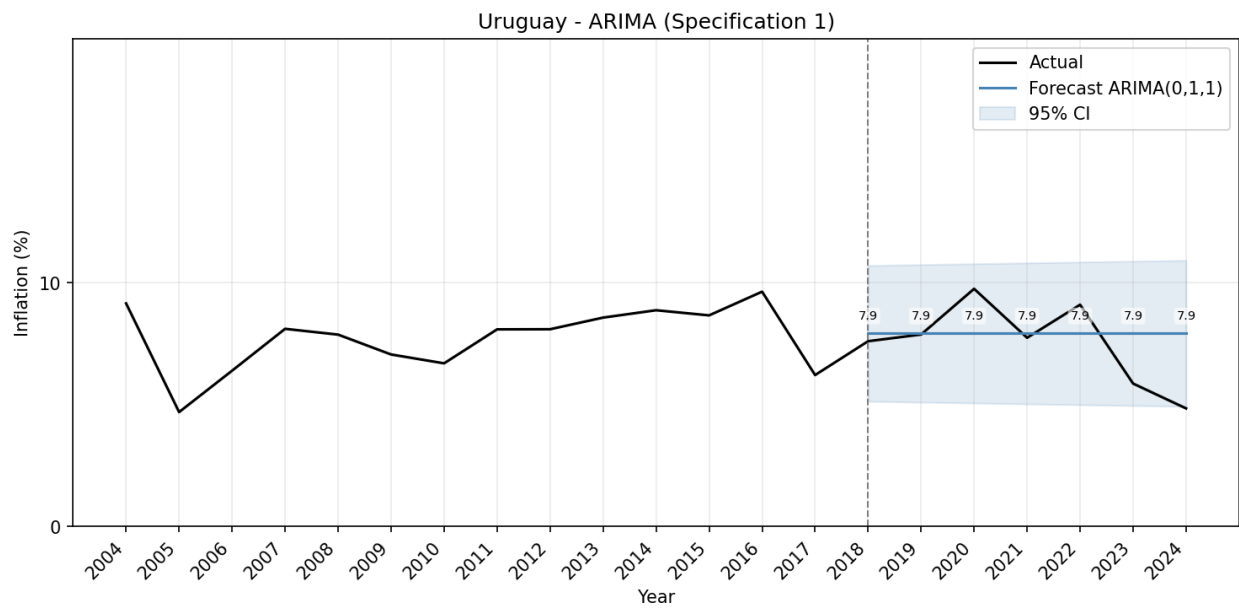


Figure 5-a

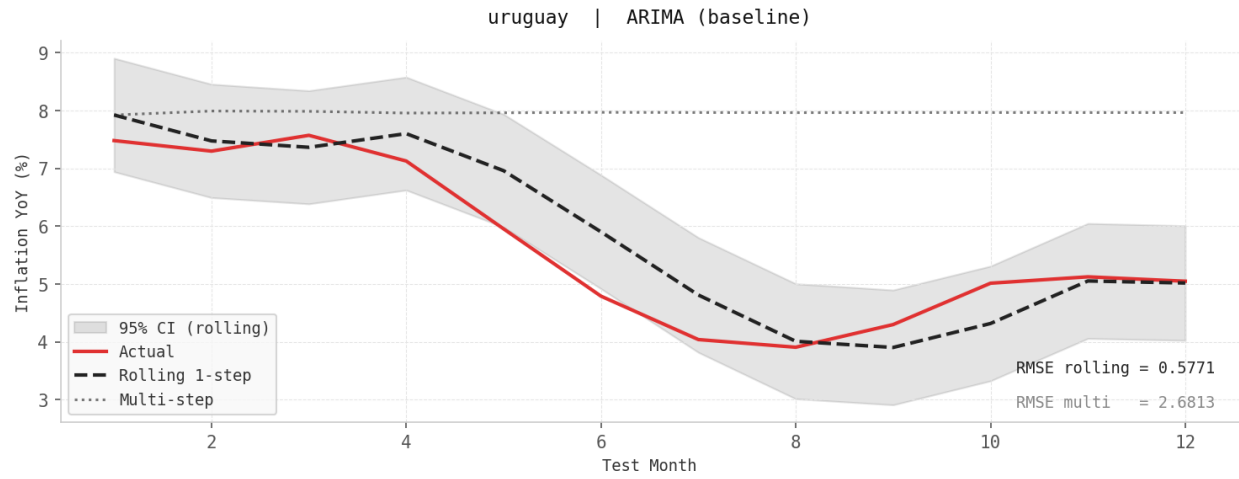


Figure 6

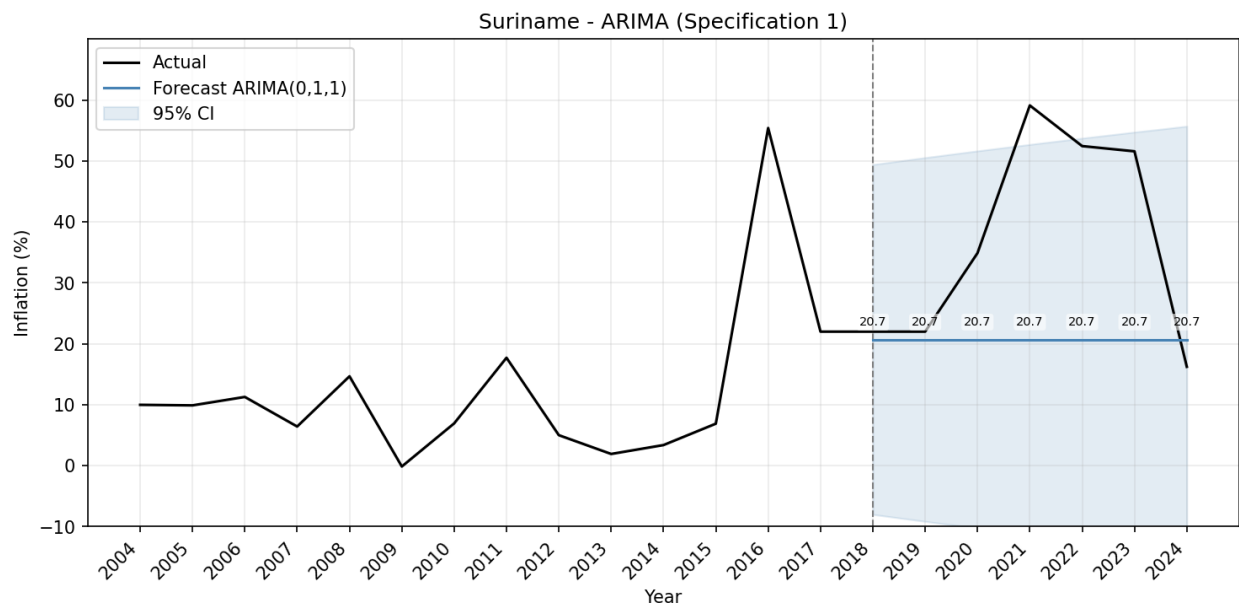


Figure 6-a

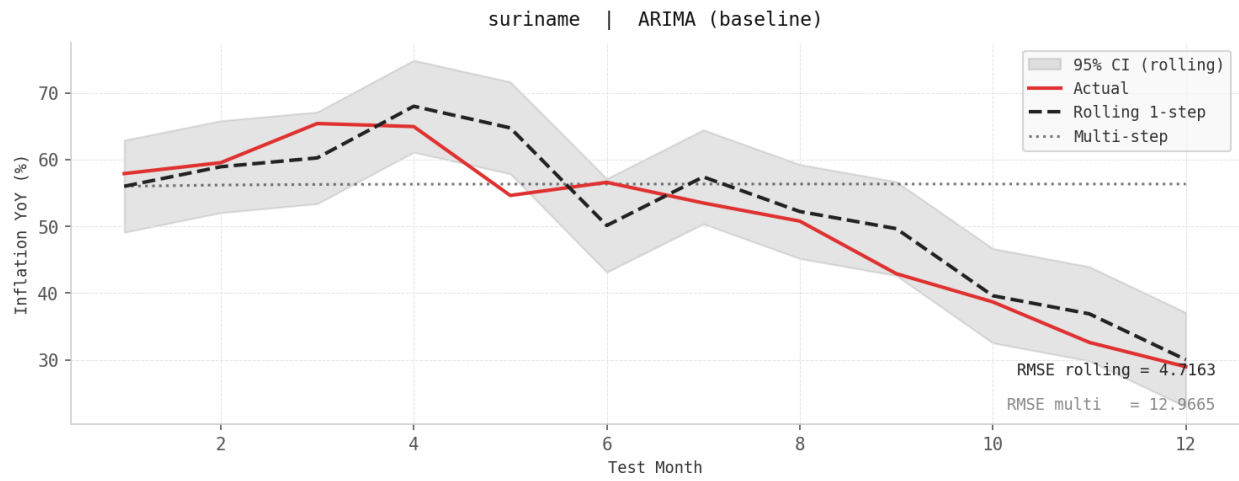
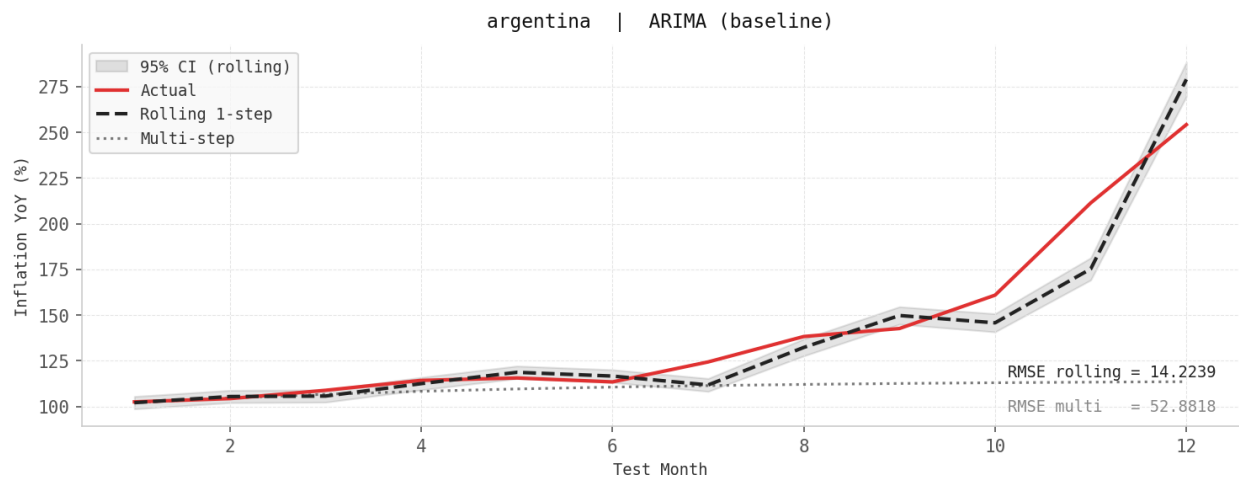


Figure 7-a



C Google Trend Keyword Selection Dictionary

C.1 CPI Channel

English	Spanish	Dutch	Persian	Portuguese (BR)	Arabic
food	alimentos	voeding	غذا	alimentos	الأغذية
non-alcoholic beverages	bebidas no alcohólicas	alcoholvrije dranken	نوشيدنى هاى غيرالكلى	bebidas não alcoólicas	المشروبات غير الكحولية
alcoholic beverages	bebidas alcohólicas	alcoholische dranken	نوشيدنى هاى الكلى	bebidas alcoólicas	المشروبات الكحولية
tobacco	tabaco	tabak	تنباكو	tabaco	التبغ
narcotics	estupefacientes	verdovende middelen	مواد مخدر	narcóticos	المخدرات
clothing	ropa	kleding	پوشاك	vestuário	الملابس
footwear	calzado	schoeisel	كفش	calçados	الأحذية
housing	vivienda	huisvesting	مسكن	habitação	السكن
water	agua	water	آب	água	المياه
electricity	electricidad	elektriciteit	برق	eletricidade	الكهرباء
gas	gas	gas	غاز	gás	الغاز
other fuels	otros combustibles	andere brandstoffen	ساير سوختها	outros combustíveis	أنواع الوقود الأخرى
furnishings	mobiliario	meubilair	اثاثيه	mobiliário	الأثاث
household equipment	equipamiento doméstico	huishoudelijke apparatuur	لوازم خانگی	equipamento doméstico	المعدات المنزلية
routine household maintenance	mantenimiento habitual del hogar	routineonderhoud van de woning	نگهدارى معمول منزل	manutenção doméstica de rotina	الصيانة المنزلية الروتينية
health	salud	gezondheidszorg	بهداشت	saúde	الصحة
transport	transporte	vervoer	حمل و نقل	transporte	النقل
information	información	informatie	اطلاعات	informação	المعلومات
communication	comunicación	communicatie	ارتباطات	comunicação	الاتصالات
recreation	ocio	recreatie	تفريح	lazer	الترفيه
sport	deporte	sport	ورزش	esporte	الرياضة
culture	cultura	cultuur	فرهنگ	cultura	الثقافة
education services	servicios educativos	onderwijs	خدمات آموزشی	serviços de educação	خدمات التعليم
restaurants	restaurantes	restaurants	رستورانها	restaurantes	المطاعم
accommodation services	servicios de alojamiento	accommodatie diensten	خدمات اسكان	serviços de hospedagem	خدمات الإقامة
insurance	seguros	verzekeringen	بیمه	seguros	التأمين
financial services	servicios financieros	financiële diensten	خدمات مالی	serviços financeiros	الخدمات المالية
personal care	cuidado personal	persoonlijke verzorging	مراقبت شخصی	cuidados pessoais	العناية الشخصية
social protection	protección social	sociale bescherming	حمايت اجتماعى	proteção social	الحماية الاجتماعية

C.2 Currency and Investment Channel

English	Spanish	Dutch	Persian	Portuguese (BR)	Arabic
dollar price today	precio del dólar hoy	de huidige dollarkoers	قیمت دلار امروز	preço do dólar hoje	سعر الدولار اليوم
exchange rate	tipo de cambio	wisselkoers	نرخ ارز	taxa de câmbio	سعر الصرف
currency converter	conversor de divisas	valuta-omzetter	مبدل ارز	conversor de moedas	محول العملات
buy dollars	comprar dólares	dollars kopen	خرید دلار	comprar dólares	شراء الدولار
currency devaluation	devaluación de la moneda	valuta devaluatie	کاهش ارزش پول	desvalorização da moeda	انخفاض قيمة العملة
exchange rate today	tipo de cambio hoy	wisselkoers van vandaag	نرخ ارز امروز	taxa de câmbio hoje	سعر الصرف اليوم
dollar price	precio del dólar	dollarkoers	قیمت دلار	preço do dólar	سعر الدولار
USD exchange rate	tipo de cambio del USD	USD-wisselkoers	نرخ ارز دلار آمریکا	taxa de câmbio do USD	سعر صرف الدولار الأمريكي
dollar	dólar	dollar	دلار	dólar	الدولار
gold	oro	goud	طلا	ouro	الذهب
gold price	precio del oro	goudprijs	قیمت طلا	preço do ouro	سعر الذهب
recession	recesión	recessie	ركود اقتصادي	recessão	الركود الاقتصادي
bankruptcy	quiebra	faillissement	ورشكستگی	falência	الإفلاس
debt	deuda	schulden	بدهی	dívida	الديون
financial crisis	crisis financiera	financiële crisis	بحران مالی	crise financeira	الأزمة المالية
invest abroad	invertir en el extranjero	in het buitenland beleggen	سرمایه‌گذاری در خارج	investir no exterior	الاستثمار في الخارج
safe investment	inversión segura	veilige belegging	سرمایه‌گذاری امن	investimento seguro	الاستثمار الآمن

C.3 Labor Market Channel

English	Spanish	Dutch	Persian	Portuguese (BR)	Arabic
unemployment	desempleo	werkloosheid	بيكارى	desemprego	البطالة
unemployment rate	tasa de desempleo	werkloosheidspercentage	نرخ بیکاری	taxa de desemprego	معدل البطالة
jobs	puestos de trabajo	banen	مشاغل	empregos	الوظائف
job vacancies	ofertas de empleo	vacatures	فرصت‌های شغلی	vagas de emprego	الوظائف الشاغرة
find a job	encontrar trabajo	een baan vinden	پیدا کردن شغل	encontrar um emprego	البحث عن وظيفة
job search	búsqueda de empleo	zoeken naar werk	جستجوی شغل	procura de emprego	البحث عن عمل
CV	CV	cv	رزومه	currículo	السيرة الذاتية
resume	currículum	—	سوابق کاری	—	—
unemployment benefits	prestaciones por desempleo	werkloosheidsuitkering	مزایای بیکاری	subsídio de desemprego	إعانات البطالة
how to get a job	cómo conseguir un trabajo	hoe vind je een baan	چگونه شغل پیدا کنیم	como conseguir um emprego	كيفية الحصول على وظيفة
layoff	despido	ontslag	اخراج	demissão	التسريح من العمل

C.4 Financial Stress Channel

English	Spanish	Dutch	Persian	Portuguese (BR)	Arabic
inflation	inflación	inflatie	تورم	inflação	التضخم
interest rate	tipo de interés	rentevoet	نرخ بهره	taxa de juros	سعر الفائدة
loan	préstamo	lening	وام	empréstimo	القرض
credit	crédito	krediet	اعتبار	crédito	الائتمان
debt relief	condonación de deuda	schuldsverlichting	تخفيف بدهی	alívio da dívida	تخفيف عبء الديون
cash assistance	ayuda económica	financiële steun	کمک نقدي	auxílio financeiro	المساعدة النقدية
defer payment	aplazamiento de pagos	uitstel van betaling	تعويق پرداخت	adiamento de pagamento	تأجيل السداد
financial crisis	crisis financiera	financiële crisis	بحران مالی	crise financeira	الأزمة المالية
bank run	pánico bancario	bankrun	هجوم به بانک	corrida aos bancos	الاندفاع على البنوك
default	impago	wanbetaling	نكول	inadimplência	التخلف عن السداد
mortgage	hipoteca	hypotheek	وام مسكن	hipoteca	الرهن العقاري
utility bill	factura de servicios públicos	energierekening	قبض خدمات عمومی	conta de serviços públicos	فواتير الخدمات العامة
what is inflation	qué es la inflación	wat is inflatie	تورم چیست	o que é inflação	ما هو التضخم
cost of living	coste de la vida	kosten van levensonderhoud	هزینه زندگی	custo de vida	تكلفة المعيشة
subsidy	subsídio	subsidie	یارانه	subsídio	الإعانة